

PUSHDOWN AUTOMATA (PDA)

A Pushdown automaton is a way to implement a Context-free grammar in a similar way we design DFA for a regular grammar. A DFA Can remember a finite amount of information, but a PDA Can remember an infinite amount of information.

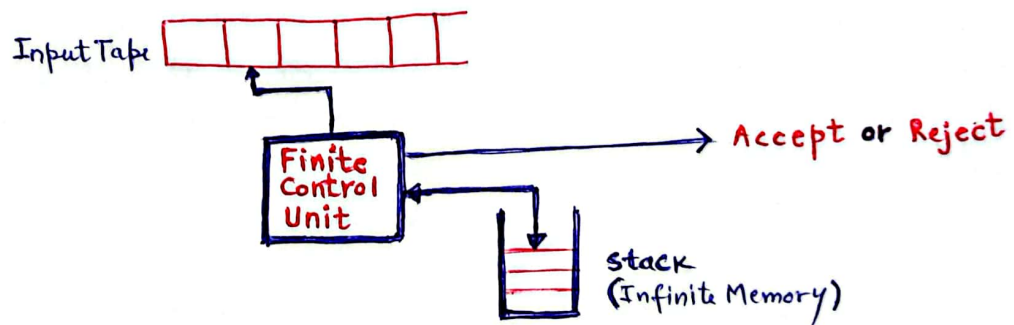
Basically a pushdown automaton is

Finite State Machine + Stack

A pushdown automaton has three Components :

- an input tape
- a Control Unit
- a Stack with Infinite size

A PDA may or may not read an input symbol, but it has to read the top of the stack in every transition.



A PDA is formally described as a 7-tuple $(Q, \Sigma, \Gamma, q_0, Z_0, F, \delta)$:

Q : Finite Set of States

Σ : Finite Set of ^{Input} alphabet

Γ : Finite Set of Stack alphabet

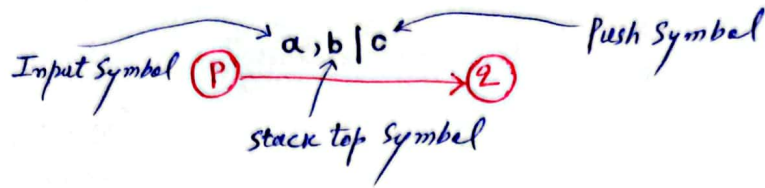
q_0 : the initial state ($q_0 \in Q$)

Z_0 : the initial stack symbol ($Z_0 \in \Gamma$)

F : Finite Set of final States ($F \subseteq Q$)

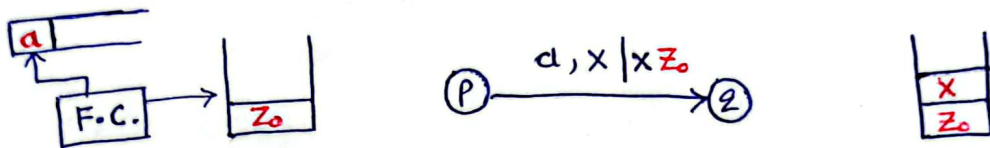
δ : is transition function : $Q \times (\Sigma \cup \{\Lambda\}) \times \Gamma \rightarrow Q \times \Gamma^*$

Transition from one state to other :



Push Instruction :

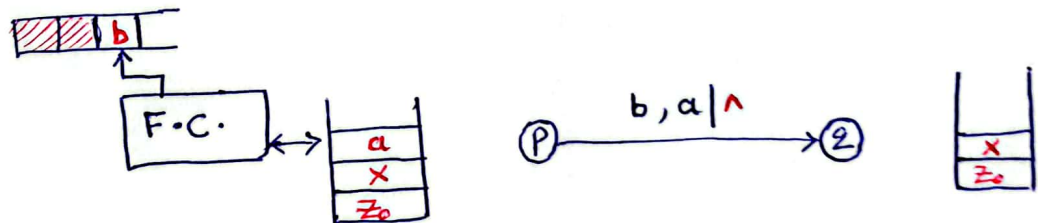
Suppose we have z_0 at the top of the stack, and we want to push x on the stack, we will write



This means at state P , if we encounter an input symbol ' a ' and there is ' z_0 ' on the top of the stack, we push ' x ' on the top of the stack and we will move to state Q .

Pop Instruction :

Suppose we have some information on the stack, where a is at the top of the stack and we want to pop a from the stack on reading b from the input tape ;



EXAMPLE:

Construct a PDA for the following language:

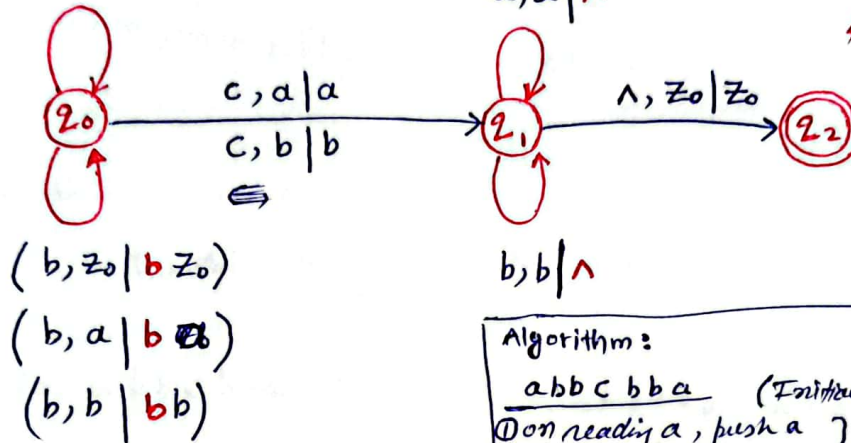
$$L = \{wcw^R \mid w \in \{a,b\}^*\}$$

CFG

$$S \rightarrow aSa \mid bSb \mid c$$

Transition Diagram

$(a, b \mid ab)$
 $(a, a \mid aa)$
 $(a, z_0 \mid az_0)$



Formally:

$$M = (Q, \Sigma, \Gamma, q_0, Z_0, F, \delta)$$

$$Q: \{q_0, q_1, q_2\}$$

$$\Sigma: \{a, b\}$$

$$\Gamma: \{a, b, z_0\}$$

$$q_0: q_0 \quad (q_0 \in Q)$$

$$Z_0: Z_0 \quad (Z_0 \in \Gamma)$$

$$F: \{q_2\}$$

$$\delta: Q \times (\Sigma \cup \{\Lambda\}) \times \Gamma \rightarrow Q \times \Gamma^*$$

Algorithm:

$abbc bba$ (Initially push z_0 on the stack) everytime
 ① on reading a , push a
 on reading b , push b } till c

② on reading c , ignore it

③ on reading a , if a is on the top of stack

on reading b , pop b if b is on the top of the stack

 $\delta:$

$$\delta(q_0, a, z_0) = (q_0, az_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, z_0) = (q_0, bz_0)$$

$$\delta(q_0, b, a) = (q_0, ba)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, c, a) = (q_1, a)$$

$$\delta(q_0, c, b) = (q_1, b)$$

 $q_1:$

$$q_1: \begin{cases} \delta(q_1, a, a) = (q_1, \Lambda) \\ \delta(q_1, b, b) = (q_1, \Lambda) \\ \delta(q_1, \Lambda, z_0) = (q_2, z_0) \end{cases}$$

ACCEPTING A WORD THROUGH PDA:

The turnstile notation is used for connecting pairs of ID's that represent one or many moves of a PDA.

A transition can be mathematically represented by the following turnstile notation:

$$(P, aW, T_h) \vdash (Q, w, ab)$$

Suppose we have:

Input string = a b b c b b a

$$(q_0, a b b c b b a, z_0) \vdash (q_0, b b c b b a, a z_0)$$

PDA moves

(Instantaneous Description)
(Pushdown Automaton Case)

Same as Derivation in CFG

$$S \Rightarrow d A B$$

$$\Rightarrow a a B$$

$$\Rightarrow a a b$$

$$S \xRightarrow{*} a a b$$

$$\vdash (q_0, b c b b a \epsilon, b a z_0)$$

$$\vdash (q_0, c b b a \epsilon, b b a z_0)$$

$$\vdash (q_1, \overbrace{b b a \epsilon}^{\text{ignore c}}, b b a z_0)$$

ignore c

$$\vdash (q_1, \overbrace{b a \epsilon}^{\text{ignore c}}, b a z_0)$$

$$\vdash (q_1, \overbrace{a \epsilon}^{\text{ignore c}}, a z_0)$$

$$\vdash (q_1, \epsilon, z_0)$$

$$\vdash (q_2, \wedge, z_0)$$

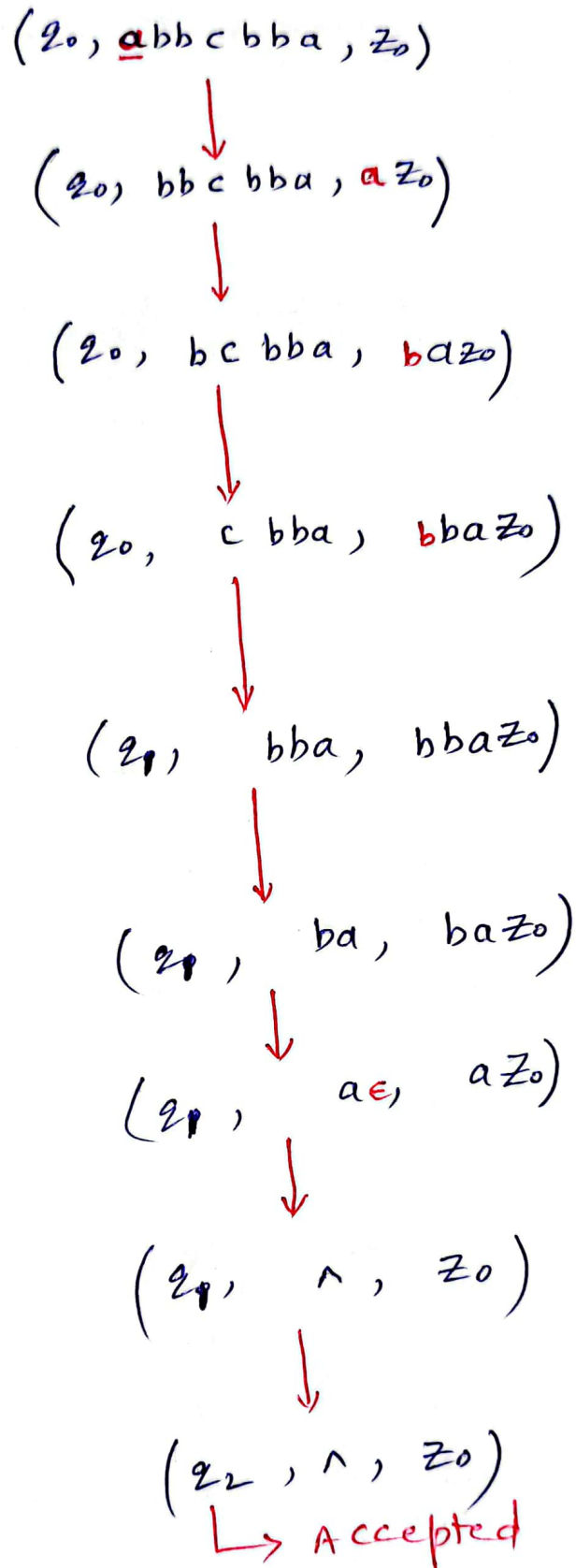
ACCEPT

OR

$$(q_0, a b b c b b a, z_0) \vdash^* (q_2, \wedge, z_0)$$

Zero or more moves of PDA

Computation Tree for the PDA with input $abbcbba$ 5



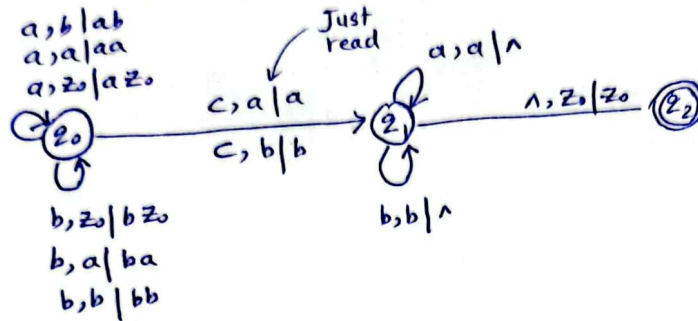
TRANSITION DIAGRAM AND TRANSITION TABLE

There are two forms to describe PDA:

- 1) Transition Diagram (more suitable description)
- 2) Transition Table

$$L = w c w^R$$

Transition Diagram:



Transition Table for $w c w^R$:

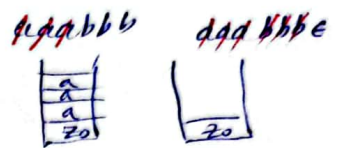
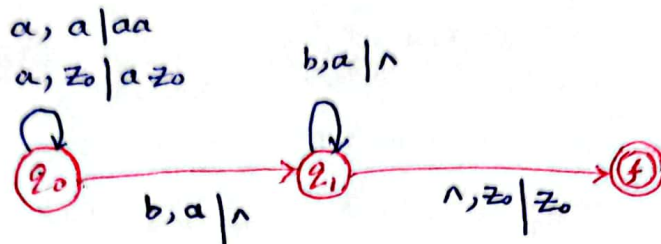
State	Input	Stack Symbol	Move(s)
q_0	a	z_0	$(q_0, a z_0)$
q_0	a	a	(q_0, aa)
q_0	a	b	(q_0, ab)
q_0	b	z_0	$(q_0, b z_0)$
q_0	b	a	(q_0, ba)
q_0	b	b	(q_0, bb)
q_0	c	a	(q_1, a)
q_0	c	b	(q_1, b)
q_1	a	a	$(q_1, \wedge)$
q_1	b	b	$(q_1, \wedge)$
q_1	$\wedge$	z_0	$(\underline{q_2}, z_0)$ (Accept)

Example:

Construct PDA for the following

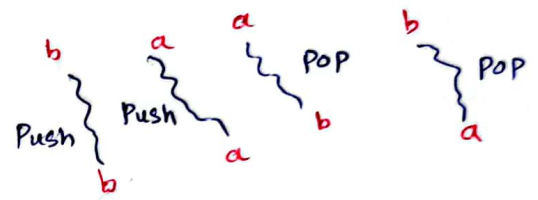
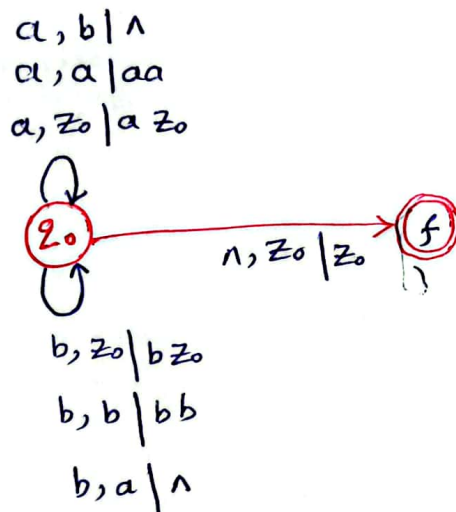
$$L = \{ a^n b^n \mid n \geq 1 \}$$

Sol:

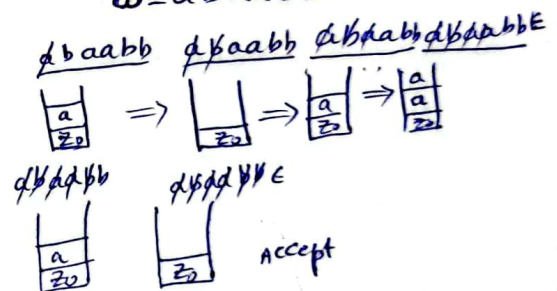


Example:

PDA for $L = \{ w \mid n_a(w) = n_b(w) \}$



$w = a b a a b b$



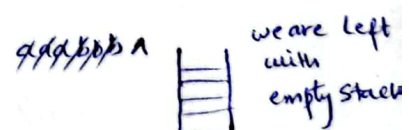
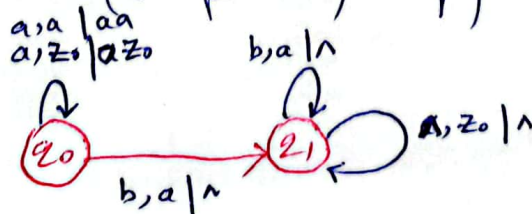
NOTE: Two representation of PDA

① Accepted by final State

② Accepted by Empty Stack

In Example above of $L = a^n b^n$, this is accepted by final State version

The other version (accepted by Empty stack) of this language is:



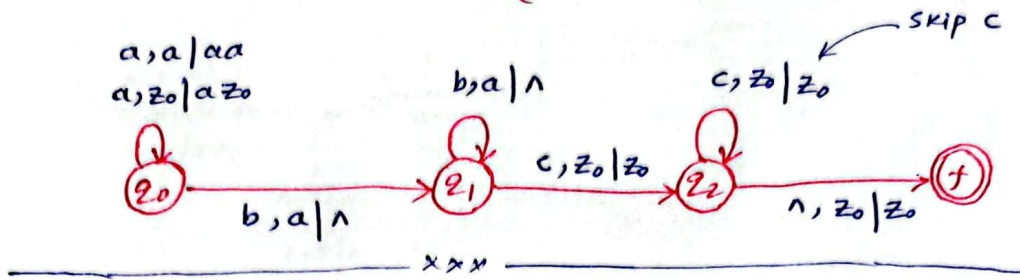
In above case we are left with



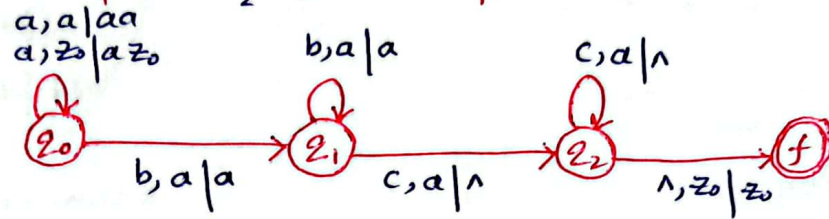
EXAMPLE :

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PDA for $L_1 = \{a^n b^n c^m \mid n, m \geq 1\}$

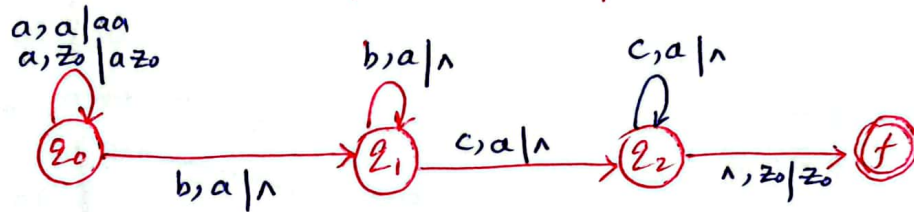


PDA for $L_2 = \{a^n b^m c^n \mid m, n \geq 1\}$



PDA for $L_3 = \{a^{m+n} b^m c^n \mid m, n \geq 1\}$

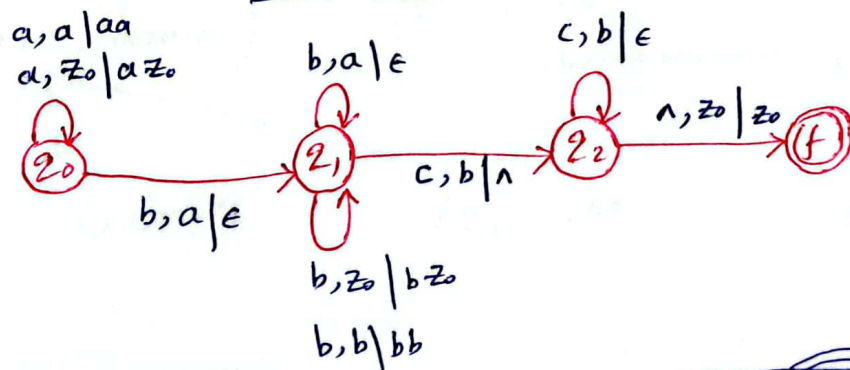
$w = \underline{aaabbc}$



PDA for $L_4 = \{a^n b^{m+n} c^m \mid m, n \geq 1\}$

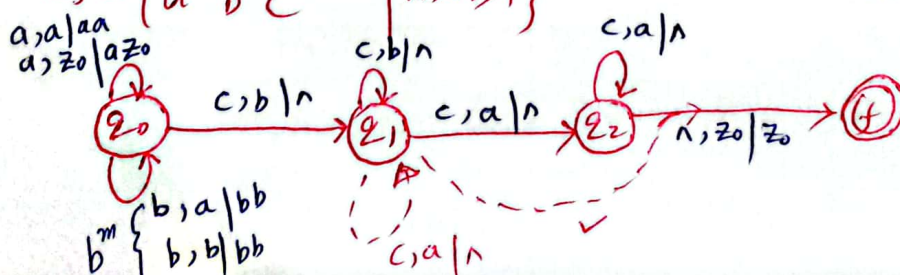
$L_4 = \underline{a^n} \underline{b^n} \underline{b^m} \underline{c^m}$

$\underline{aa} \underline{bbbb} \underline{cc}$



$L_5 = \{a^n b^m c^{m+n} \mid m, n \geq 1\}$

$\underline{a^n} \underline{b^m} \underline{c^{m+n}}$



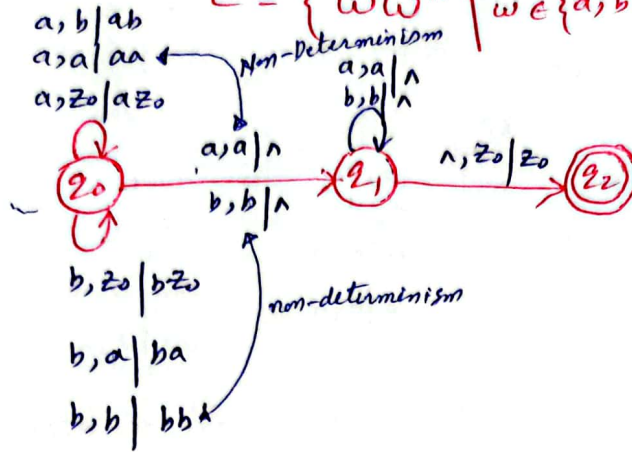
EXAMPLE:

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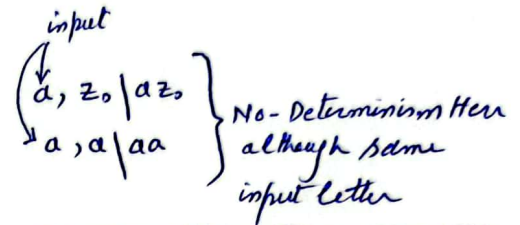
NPDA

$$S \rightarrow asa | bsb | \Lambda$$

$$L = \{ ww^R \mid w \in \{a, b\}^* \}$$



EVENPAL



In PDA

$\left. \begin{array}{l} a, a | aa \\ a, a | \Lambda \end{array} \right\}$ Same input letter and same stack alphabet.

This is the case of NonDeterminism

At q_2 , there is no outgoing edge - In case of PDA, there is no issue of a state having no outgoing edge

no such transition present here

$(q_0, aaaa, z_0)$

(q_0, aaa, az_0)

NC

Centre

(q_0, aa, aaz_0)

(q_1, aa, z_0)

X

No Centre

$(q_0, a, aaaa z_0)$

$(q_1, a, a z_0)$

NC

$(q_0, \Lambda, aaaa z_0)$

$(q_1, \Lambda, aa z_0)$

X

(q_1, Λ, z_0)

(q_2, Λ, z_0)

Accept

DPDA is not possible

Also $DPDA \subset NPDA$

DPDA has less power than NPDA.

EXAMPLE

PDA for PALINDROME (odd + even = both)

NPDANo version
of DPA exists
 $a, b | ab$
 $a, a | aa$
 $a, z_0 | az_0$
 q_0
 $b, z_0 | bz_0$
 $b, a | ba$
 $b, b | bb$

(λ-transition)

 $λ, a | a$
 $λ, b | b$
 $a, a | a$
 $a, b | b$
 $b, a | a$
 $b, b | b$
 $a, a | λ$
 q_1
 $b, b | λ$
 $λ, z_0 | z_0$
 q_2
(YES) odd $ab \underline{a} ba$ (YES) even $abba$ ~~λ-transition~~

Computation Tree for the PDA with baab

 $(q_0, baab, z_0)$ (q_0, aab, bz_0) (q_0, ab, abz_0) (q_1, ab, bz_0) (q_1, aab, bz_0) $(q_0, b, aabz_0)$ (q_1, b, abz_0) (q_1, ab, abz_0) $(q_0, λ, baabz_0)$ $(q_1, λ, aabz_0)$ $(q_1, b, aabz_0)$ $(q_1, λ, baabz_0)$ The Sequence of moves that
Leads to acceptance $(q_0, baab, z_0) \vdash (q_0, aab, bz_0)$ $\vdash (q_0, ab, abz_0)$ $\vdash (q_1, ab, abz_0)$ $\vdash (q_1, b, bz_0)$ $\vdash (q_1, λ, z_0)$ $\vdash (q_2, λ, z_0)$ Acceptance

① ODD Length, No Center

baab

NC (read and push)

baab ② ODD Length + Center
just readodd length
Center - YESbaab ③ Even Length
+ Center (λ-transition)

odd case (C)

baab (C) - YES
Even lengthλ-move
(no read
no push) (q_1, b, bz_0) $(q_1, λ, z_0)$ $(q_2, λ, z_0)$

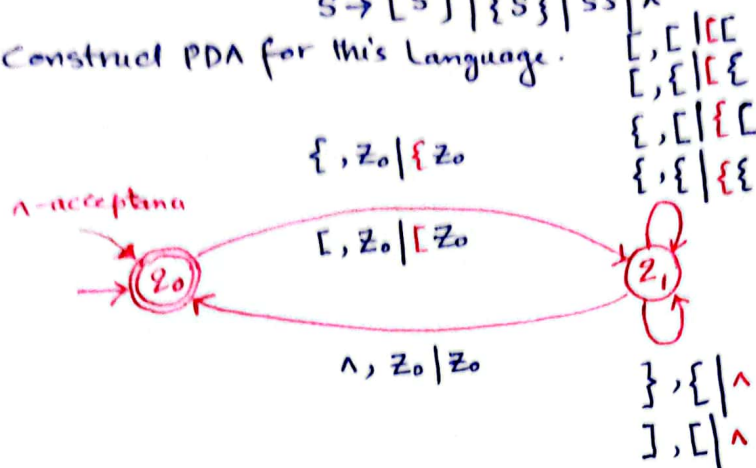
(ACCEPT)

EXAMPLE :

Consider the language L of all balanced strings involving two types of brackets: $\{ \}$ and $[]$. L is the Language generated by CFG :

$$S \rightarrow [S] \mid \{S\} \mid SS \mid \wedge$$

Construct PDA for this Language.



string: $\{ [] \} []$ (balanced)

$$(q_0, \{ [] \} [], z_0) \vdash (q_1, [] \} [], \{ z_0)$$

$$\vdash (q_1,] \} [], [\{ z_0)$$

$$\vdash (q_1, \} [], \{ z_0)$$

$$\vdash (q_1, [], z_0)$$

$$\vdash (q_1,], [z_0)$$

$$\vdash (q_1, \wedge, z_0)$$

$$\vdash (q_0, \wedge, z_0) \text{ accept}$$

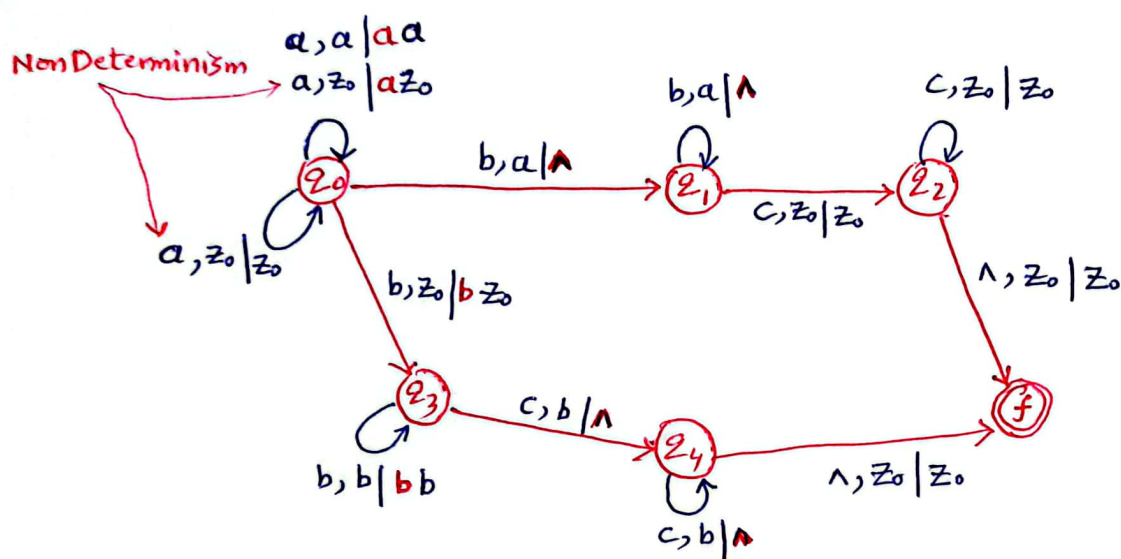
PDA for $a^i b^j c^k$, where $i=j$ or $j=k$

Sol:-

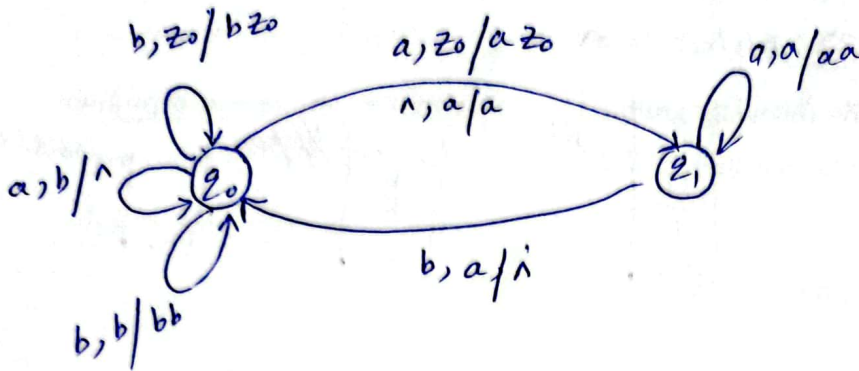
$a^i b^j c^k \mid i=j \text{ or } j=k$ means $a^n b^n c^m \mid m, n \geq 1$ or $a^m b^n c^n \mid m, n \geq 1$

$a^n b^n c^m \mid m, n \geq 1$ means $a^n b^n c^* \mid n \geq 1$

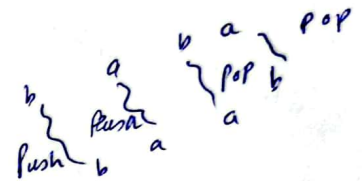
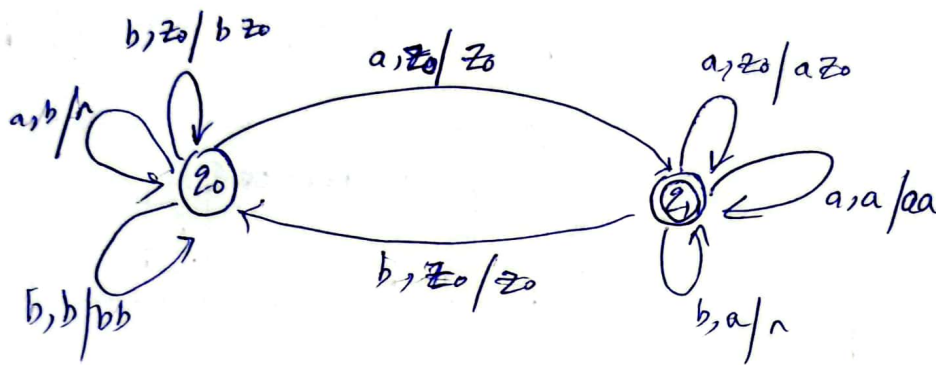
$a^m b^n c^n \mid m, n \geq 1$ means $a^* b^n c^n \mid n \geq 1$



$$L = \{x \in \{a,b\}^* \mid n_a(x) > n_b(x)\}$$



with no λ -transitions



$$\begin{aligned} \delta(q_0, a, z_0) &= (q_1, z_0) \\ \delta(q_0, b, z_0) &= (q_1, bz_0) \end{aligned}$$

$$\begin{aligned} \delta(q_0, a, b) &= (q_0, ^) \\ \delta(q_0, b, b) &= (q_0, bb) \end{aligned}$$

$$\begin{aligned} \delta(q_1, a, z_0) &= (q_1, az_0) \\ \delta(q_1, b, z_0) &= (q_0, z_0) \end{aligned}$$

$$\begin{aligned} \delta(q_1, a, a) &= (q_1, aa) \\ \delta(q_1, b, a) &= (q_1, ^) \end{aligned}$$

$$(q_0, abbabaa, z_0) \vdash (q_1, bbabaa, z_0)$$

$$\vdash (q_0, babaa, bz_0)$$

$$\vdash (q_0, abaa, bz_0)$$

$$\vdash (q_0, baa, z_0)$$

$$\vdash (q_0, aa, bz_0)$$

$$\vdash (q_0, a, z_0)$$

$$\vdash (q_1, ^, z_0) \text{ accept}$$